

Question	Answer	Marks
2(c)(iii)	<i>any four from</i> 1 (reversibly) binds / attaches / AW, to, allosteric site / site other than active site ; 2 (which) changes, shape / tertiary structure / 3-D structure, of <u>active site</u> ; A active site distorted I protein structure 3 substrate / ATP, cannot, enter / bind / fit / AW, to active site ; A active site no longer complementary to substrate A enzyme substrate / ES, <u>complexes</u> cannot form A ESCs cannot form I ATP / substrate, cannot bind to enzyme <i>without a link to <u>active site</u></i> 4 no / less, hydrolysis of ATP A breakdown or no / less, energy released ; I no energy, synthesised / created / produced 5 Na⁺ not moved, out <u>and</u> K⁺ not moved in ; I active transport, stops / decreases	4

Question	Answer	Marks
2(d)(i)	<i>allow systole for contraction and diastole for relaxation</i> <i>bicuspid valve or mitral valve for (left) atrioventricular</i> <i>aortic valve for semi-lunar valves</i> <i>any four from (max 3 if whole response based on right side of heart)</i> <i>before atrial contraction / during relaxation of the left atrium and left ventricle</i> 1 atrioventricular valve, opens / is open A <i>following atrial contraction</i> or blood trickling into ventricle / some blood enters ventricle ; 2 atrial contraction, blood flow to ventricles / ventricles fill (with blood) or atrial contraction then ventricular contraction ; <i>ventricular contraction</i> 3 bicuspid valve closes <u>and</u> semi-lunar valve opens ; R <i>if occurs before ventricular contraction</i> 4 blood flows into aorta ; R <i>if states 'from atrium' or 'then to lungs'</i> R <i>if occurs before ventricular contraction</i> 5 <i>ref. to</i> atrium in relaxation during ventricular contraction ; <i>pressure changes</i> 6 contraction of, atrium / ventricle, increases pressure (of that chamber) or <i>ref. to</i> (blood) pressure differences to cause opening or closing of valves ; e.g. pressure in atrium greater than in ventricle so atrioventricular valve opens pressure in ventricle greater than aorta so semilunar valves open pressure in ventricle greater than atrium so bicuspid valve closes	4

Question	Answer	Marks
2(d)(ii)	I <i>ref. to</i> fibrillation / cardiac cycle rhythm <i>any three from</i> 1 more powerful contraction of (cardiac) muscle / increased ability for (cardiac) muscle (A cardiac cells) to contract ; A stronger contraction / contract strongly / increased contractility I contracts more / increased contraction 2 blood (pumped) at higher<u>e</u>r pressure ; I blood at high pressure 3 more force to overcome resistance (in blood vessels) ; 4 more blood reaches lungs to obtain oxygen (per unit time) / more oxygen reaches (rest of) body / tissues (per unit time) (in blood) ; <i>allow idea of efficient delivery of oxygen</i> A more oxygenated blood can be delivered to heart, muscle / tissue AW 5 less fatigue / increased energy / increased mobility / AW ;	3